

Design Calculation Sheet - VBF-T1R1-CALC-01

Energy density chain

Item	Value	Formula	Source
1C capacity	5.6505 Ah	1C_discharge protocol	cell/r3_l_1c.json
Energy	19.9673 Wh	integral $V \times I \, dt$	cell/r3_l_energy.json
Stack mass	0.036044 kg	sum layer thickness x (1-porosity) x density x area	cell/r3_l_energy.json (recomputed match 1e-12)
ED gravimetric	553.977 Wh/kg	energy / mass	cell/r3_l_energy.json
ED volumetric	905.983 Wh/L	energy / volume	cell/r3_l_energy.json

N/P and mass

Item	Value	Formula
N/P formula caliber	0.7520	$(33133 \times 0.60 \times 120 \, \mu\text{m}) / (63104 \times 0.665 \times 75.6 \, \mu\text{m})$
N/P usable-window caliber	~1.0	anode 0.860 vs cathode 0.647 of full capacity at 2.5-4.2 V
Positive layer	16.8422 g	$75.6 \, \mu\text{m} \times 0.665 \times 3262 \times 0.1027 \, \text{m}^2$
Negative layer	12.2525 g	$120 \, \mu\text{m} \times 0.60 \times 1657 \times 0.1027 \, \text{m}^2$
Al CC / Cu CC	2.2183 / 4.6010 g	$8 \, \mu\text{m} \times 2700 / 5 \, \mu\text{m} \times 8960 \times 0.1027 \, \text{m}^2$
Separator	0.1297 g	$6 \, \mu\text{m} \times 0.53 \times 397 \times 0.1027 \, \text{m}^2$

Process parameters

Parameter	Value	Formula
Positive areal density	163.994 g/m ² = 16.399 mg/cm ²	thickness x (1-porosity) x density
Negative areal density	119.304 g/m ² = 11.930 mg/cm ²	thickness x (1-porosity) x density
Positive compaction density	2169.2 kg/m ³ = 2.169 g/cm ³	density x (1-porosity) / 1000
Negative compaction density	994.2 kg/m ³ = 0.994 g/cm ³	density x (1-porosity) / 1000
Electrolyte fill	9.3842 g	pore volume 7.820 cm ³ x 1.2 g/cm ³ x 1.0
Formation	0.1C CC to 4.2 V, 25 degC, 2 cycles	design recommended value